◇ Python Variables – Frequently Asked Interview Questions

1. What is a variable in Python?

A variable is a name that refers to an object stored in memory.

2. Is Python statically typed or dynamically typed?

Python is dynamically typed, meaning type is decided at runtime.

3. How does Python handle variable memory allocation?

Python allocates memory automatically and variables store references to objects, not values.

4. Can a variable change its data type in Python?

Yes, a variable can refer to objects of different data types at different times.

5. What is the difference between variable and object in Python?

A variable is a reference, while an object is the actual data in memory.

6. What happens when multiple variables point to the same object?

All variables refer to the same memory object; changes affect all references (for mutable objects).

7. What is variable shadowing in Python?

When a local variable overrides a global variable with the same name.

8. What is the scope of a variable in Python?

Scope defines where a variable can be accessed (local, global, enclosing, built-in).

9. What is the difference between local and global variables?

Local variables are defined inside a function, global variables are defined outside all functions.

10. What is the use of the global keyword?

It allows a function to modify a global variable.

11. What is the use of the nonlocal keyword?

It allows modifying a variable from the nearest enclosing (non-global) scope.

12. Are variables in Python passed by value or reference?

Python uses call by object reference (call by sharing).

13. What is the difference between assignment (=) and comparison (==)?

= assigns a value, == compares values for equality.

14. What is the purpose of the id() function?

It returns the unique identity (memory address) of an object.

◇ Python Data Types – Frequently Asked Interview Questions

15. What are the built-in data types in Python?

int, float, complex, str, list, tuple, set, dict, bool, NoneType, bytes.

16. What is the difference between mutable and immutable data types?

Mutable data types can be changed, immutable data types cannot be changed after creation.

17. Which Python data types are immutable?

int, float, bool, str, tuple, frozenset, NoneType.

18. Which Python data types are mutable?

list, dict, set, bytearray.

19. What is the difference between list, tuple, and set?

List: ordered, mutable

Tuple: ordered, immutable

Set: unordered, mutable, unique elements

20. What is the difference between list and array?

Lists can store mixed data types, arrays store same data type elements.

21. What is the difference between None and 0?

None means no value, 0 is a numeric value.

22. What is the difference between None and False?

None represents absence of value, False represents a boolean condition.

23. What is NoneType?

It is the data type of None.

24. What is the difference between == and is?

== checks value equality, is checks object identity.

25. What is type casting in Python?

Converting one data type into another.

26. What is implicit vs explicit type conversion?

Implicit is done automatically, explicit is done manually by the programmer.

27. How do you check the type of a variable?

Using the type() function.

28. What is the difference between int and float?

int stores whole numbers, float stores decimal numbers.

29. What is the bool data type derived from?

It is derived from the integer type (True = 1, False = 0).

30. Are strings mutable or immutable in Python?

Strings are immutable.

◇ Python Naming Rules – Frequently Asked Interview Questions

31. What are the rules for naming variables in Python?

Must start with a letter or underscore, cannot start with a number, cannot be a keyword.

32. Can a variable name start with a number?

No.

33. Can Python variable names contain special characters?

No, only letters, numbers, and underscore are allowed.

34. Are Python variable names case-sensitive?

Yes.

35. Can we use Python keywords as variable names?

No.

36. What is PEP 8 naming convention?

It is Python’s official style guide for writing readable code.

37. What is snake_case in Python?

A naming style using lowercase letters with underscores.

38. What is the difference between _var, __var, and __var__?

_var: protected (convention)

__var: name mangling

__var__: special/magic variables

39. What are magic (dunder) variables in Python?

Variables with double underscores used internally by Python.

40. Why should meaningful variable names be used?

To improve code readability, maintainability, and clarity.